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A Feature Selection-Based Ensemble Machine Learning Method for Predicting Chronic Kidney Cancer

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***Abstract*— Chronic kidney cancer (CKC) is a real-world example of a crucial class of frontiers that are behind basic medical diagnostics and have a significant worldwide impact. For this reason, we need to evaluate more sophisticated prediction algorithms. In order to balance the previously proven CKC detection, this research has presented a novel strategy that combines ensemble approaches with robust machine learning. We employ nine alternative algorithms three from each of Random Forest, Decision Tree, and Light Gradient Boosting Machine in addition to the standard predictive models for this specific stacking ensemble. We obtained an accuracy of 99.19% after preprocessing the data and using collaboration for feature selection. Our model performs with excellent predictive power, similar to Random Forests, but with the ease of de-biasing and good interpretability of Decision Trees. Additionally, as a mixed category, the ensemble performs better than any one model on all performance criteria (precision, recall, and F1 score). In order to enhance individual risk prediction of CKC and intervention execution, this adaptable framework is therefore ideal for use in clinical settings. Let's put it this way: our study is a significant step toward improving CKC diagnosis, and it provides a foundation for the broader application of ensemble models of learning in healthcare, which might revolutionize patient care in all specializations.**

Keywords*— *Chronic Kidney Cancer (CKC), Machine Learning, Feature Selection, Stacking Ensemble Technique, Random Forest, Decision Tree, Hybrid Model, Predictive Analytics

I. INTRODUCTION

On an annual basis, Chronic Kidney Cancer (CKC) impacts around 10% of the global population. Furthermore, the World Health Organization (WHO) has shown an increasing prevalence of CKC in the majority of developing countries [1]. Primary victims of this disease are predominantly third-world nations, where the bulk of the population consists of individuals with poor and medium incomes [2]. Kidney failure Chronic kidney disease arises when the kidney is unable to effectively carry out its primary role, which is filtering the blood in the human body. In cases of CKC, the kidney is unable to sustain the appropriate filtration of extrinsic substances from the human bloodstream. This ultimately results in serious health complications such as nerve damage, hypertension (both high and low blood pressure), diabetes, skeletal disorders, and other diseases.

Consequently, in their later years, people with CKC experience particular adverse consequences that impair their immune system and neurological system [3]. An inherent drawback of CKC is its inability to accurately evaluate symptoms. Primary etiological factors of CKC include previous familial predisposition, hypertension, and diabetes. Accurate identification of CKC requires an early diagnosis, yet, the expense of the traditional testing method is too high for the majority of individuals in poor nations. Identification of CKC at advanced stages might result in renal failure. The sick individual must undergo kidney transplantation or continuous dialysis to sustain a normal lifestyle, which is not to mention prohibitively costly [4]. In order of importance, prior family issues, hypertension, and diabetes are related to CKC [5,6].

Artificial intelligence is required by the medical sector due to the daily growth in the number of affected individuals and the escalating rate of this crisis. This technique is thought to enable anyone to assess the likelihood of being impacted by CKC without incurring any cost. The primary challenge in employing machine learning algorithms for early diagnostic of CKC is the insufficient availability of both high-quality and large amounts of data. The presence of noisy, incomplete, imprecise, and erroneous data leads to substandard outcomes and less accurate classification. As a result, inadequate quality of information can also be viewed as a major risk when detecting CKC early on.

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In order to decrease the adverse consequences of this problem, we propose in our study domain a machine learning based prediction model for Chronic Kidney Cancer. The first step is to preprocess the gathered data. Subsequently, we integrate many machine learning algorithms into the preprocessed datasets. Next, we analyze the output of the model by employing a range of metrics.

Many research has used different classification methods to predict distinct stages of CKC. This section examines seminal studies on CKC prediction.

Islam and colleagues [7] introduced a model that utilizes the Random Forest algorithm to forecast competitive knowledge cost (CKC). Samples from a collection of 1032 patients were used in this investigation. The degree of accuracy attained by the model was 98.88%. A neural network-based framework (NN) for diagnosing chronic kidney disease was introduced by Kim and colleagues [8]. Their suggested genetic algorithm optimized the weight vectors for training the neural network. In this study, the classification task was only performed using the artificial neural network (ANN), resulting in an accuracy rate of 95.4. Qin et al.'s [9] novel method of identifying chronic renal disease has a remarkable 99.83% accuracy rate. This combined approach makes use of a perceptron-based framework to optimize the capabilities of both random forest analysis and logistic regression. It continuously attained an average accuracy of 99.83% throughout ten iterations.

In order to detect chronic renal illness, Wang et al. [10] suggested a machine learning method that makes use of the random forest, XGBoost, and ResNet algorithms. The predicted accuracy rating for the following model is 76%. The collection included data from one million physical examinations, with ages ranging from 25 to 90 Using data from Taiwan's National Health Insurance Research Database on drug use and comorbidities, Krishnamurthy et al. [11] developed a machine learning model with initial findings indicating an accuracy of 71.2%. Thereafter, the CNNs were incorporated into the model. In such a way, the accuracy subsequently improved to 90.5%.

Mohammed et al. [12] developed the accuracy measure and used multiclass statistical analysis to assess and analyze it. With support vector machine (SVM), decision tree and k-nearest neighbor's (KNN) classification algorithms, the authors achieved notable accuracy rates of 98.33%, 96.67% and 99.17%. Rady et al., provided an ML approach using a combination of MLP, RBF, SVM and PNN techniques. [13]. The method again has the best accuracy, 96.7%, so results were higher than SVM, MLP, and RBF algorithm. Vikas Chaurasia et al. also recommended a machine learning approach built using Decision Tree analysis for prediction scenarios [14] in 2018. This method achieved 93% accuracy. Xiao et al. In another study [15], nine machine learning models were used to predict CKD progression: LR, SVM, Elastic Net, XGBoost, KNN, RF, naïve Bayesian and neural networks. Using this data, the robot achieved an 80% precision using my chosen model

II. Materials and Methods

This study compared machine learning algorithms with other traditional methods for predicting kidney cancer. We try forming a combination approach that could be used to build the prediction algorithm by considering 9 models' performance, e.g., Naive Bayes (NB), Logistic regression (LR), Decision Tree (DT), k-Nearest Neighbor (k-NN), Random Forest (RF), Support Vector Machines (SVM), AdaBoost is (AB), CatBoost (CB) and Light Gradient Boosting Machine (LGBM). This was done to see how effective and dependable these algorithms were in identifying cancer.

The CKC in order predictions architecture ("Fig. 1") starts with pre-processing the data, analyzing attribute correlations to identify critical characteristics for kidney cancer identification. Subsets for testing and training are then created from the dataset. Using the training data, a variety of machine learning approaches are used to create prediction models. Several confusion measures are used to evaluate the model's performance

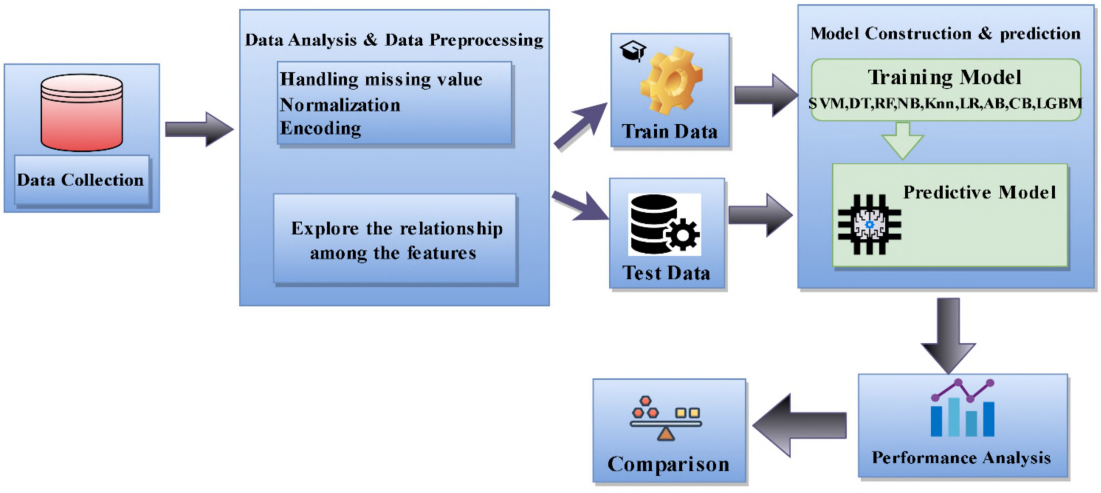


Fig.1. Workflow model for the proposed system

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A. Dataset Collection

This study was conducted using a dataset collected from CKD patients at Apollo Hospitals in India over two months in 2015.The "Chronic Kidney Disease Dataset" (available in the UCI repository) contained 400 entries with noisy and missing values [16].These patients range in age from 2 to 90 years, with 62.5% having CKD and 37.5% not having CKD. There are 24 variables in the dataset: 11 are numeric, 13 are nominal, and the 25th variable is the classification label indicating whether or not CKD is present.

B. Dataset preprocessing

One essential stage in the creation of a machine learning model is data preparation; it guarantees that raw data is cleaned up and prepared for model integration. Finding properly organized data is uncommon since datasets frequently have irregularities. Therefore, comprehensive data cleaning and restructuring are necessary prior to model deployment. This preparatory stage not only improves the data quality but is essential to maximizing the model's overall performance. The effectiveness of algorithms developed for machine learning can be seriously harmed by improper preprocessing.

C. Missing Value

The dataset displayed in "Fig. 2" has 25 columns, and the bar chart displays the quantity of missing values in each column. The y-axis displays the column names, while the x-axis ranges from 0 to 400, which represents the total amount of data points. Notably, all 400 values are included in the first column, "ID," which is completely filled up with no blanks. In comparison, there are 391 missing values in the "age" column, leaving only 9 full entries. This bar chart shows the frequency of incomplete records for almost all variables and offers a thorough overview of the data that is absent distribution.

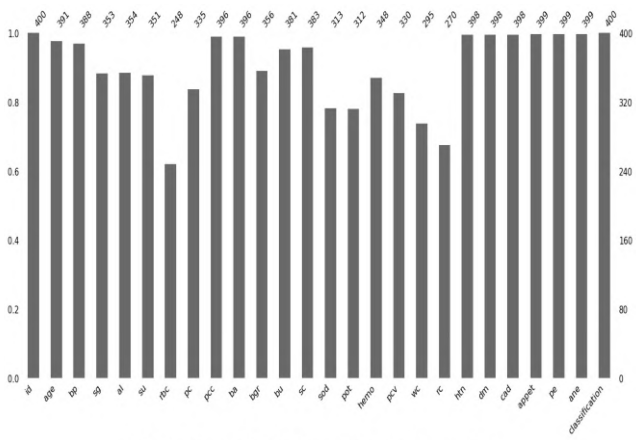


Fig.2.Graphical view of missing value

In the present research, NULL entries were replaced with predetermined values as part of the Fillna technique to handle missing data. Fillna conducts the replacement directly among the original dataset unless the `inplace` argument is specified to {False}, in which case it creates a new DataFrame. The approach first deals with missing values related to 'blood urea nitrogen', 'blood urea nitrogen concentration', 'age', 'blood pressure', 'specific gravity', 'hemoglobin', 'albumin', 'blood sugar', and 'potassium'. Then, it further treats the categorical variables that include 'packed cell volume', 'erythrocytes', 'pus cells', 'pus cell clusters', 'bacteria', 'white blood cell count', 'red blood cell count', 'hypertension', 'diabetes', 'coronary heart disease', 'appetite', 'pedal edema', and 'anemia'. This thorough Fillna method guarantees that any missing data is removed.

D. Feature Selection

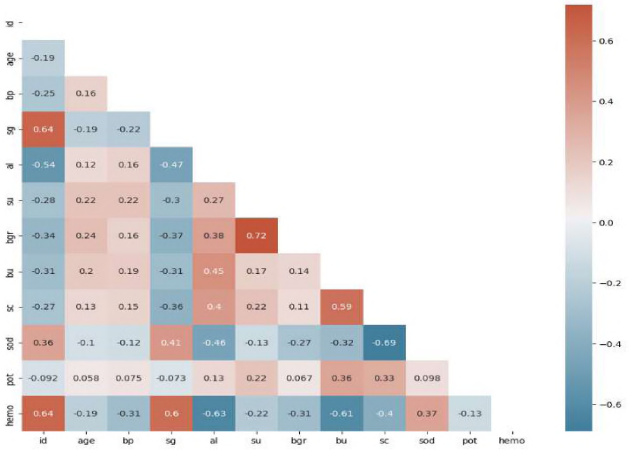
The practice of methodically locating the most pertinent attributes in a dataset is known as feature selection. By decreasing overfitting and creating model's costs, this method improves model performance. Filter-based feature selection techniques find characteristics that show greater correlation with the desired variable, independent of the particular learning methodology. To evaluate the quality of chosen features, an independent assessment measure is employed, which is not reliant on the framework for learning. In this work, mutual information [17] was used as the metric to evaluate the dependence between variables, and picking features was based on a traditional information-theoretic criterion.

A comparison of the heat map results generated with and without feature selection is shown in "Fig. 3." Before and after feature selection, correlation heat maps are compared. With color intensity indicating the strength of each association, the heat maps graphically depict the relationships between the dataset's many properties. While the right subfigure (b) displays correlations following the removal of unnecessary features using feature selection, the left subfigure (a) displays correlations with all original characteristics. The strength of the link is measured by the correlation coefficient, or r. The degree of the linear link between two variables may be measured statistically using the Pearson's inverse association coefficient, as it was named after its discoverer. Values close to -1 or +1 on the coefficient's range of -1 to +1 imply a perfect linear correlation, whether positive or negative. There is a positive correlation when both variables rise simultaneously. One measure shows a concurrent growth with the other when there is a positive association. We get a negative value when one value falls while the other grows. The exact definition of zero is the total absence of connection. This illustration demonstrates how feature selection, which only considers relevant attributes, may enhance model interpretability and perhaps prediction accuracy.

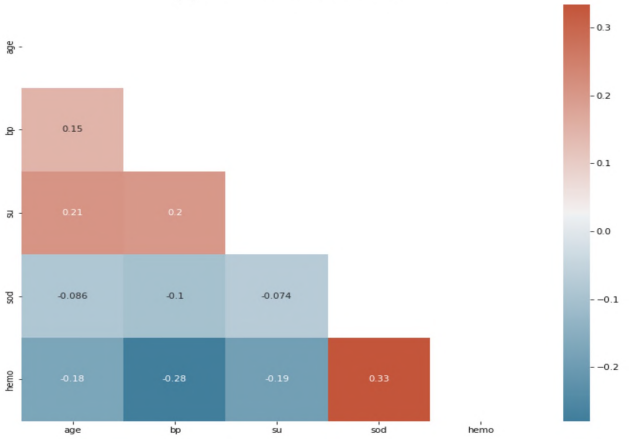
Our feature selection process significantly improved the performance of our model. We choose which features to retain based on mutual information. In essence, it identifies the elements that have a strong connection to the prediction we are attempting to make. After we select only the most useful qualities, you can observe how the significant features changed in "Fig. 3(a)" and "Fig. 3(b)". The model operates

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more quickly and doesn't overfit as much when unnecessary features are reduced. In this manner, it becomes more capable of handling new facts and less inclined to just memorize things.



(a) Before features selection



(b) After features selection

Fig.3. Comprises of feature selection (a) before features selection, (b) after features selection

"Fig. 4" depicts the relationships between variables. Equation 1 calculates the correlation coefficient, with (n) representing the values of the independent variable and (n) representing the values of the dependent variable. To comprehend how variables depend on one another, this correlation matrix is crucial.

$$r = \frac{\sum(a-\bar{a})(b-\bar{b})}{\sqrt{[\sum(a-\bar{a})^{-2}(b-\bar{b})^{-2}]}} \qquad (1)$$

Equation (1) was used to calculate both positive and negative connections between variables, and the resulting correlation matrix is shown in "Fig. 4" across every single column in the dataset. Comparably, "Fig. 4" shows the correlation unique to the selected column, and Equation (1) is likewise used to get the correlation coefficient. An extensive and focused summary of the correlations between the variables in the dataset is given by these visuals. By emphasizing characteristics that have a significant correlation with the target variable, these connections help in feature selection. This improves model performance by minimizing redundancy and guaranteeing the selection of the most informative qualities.

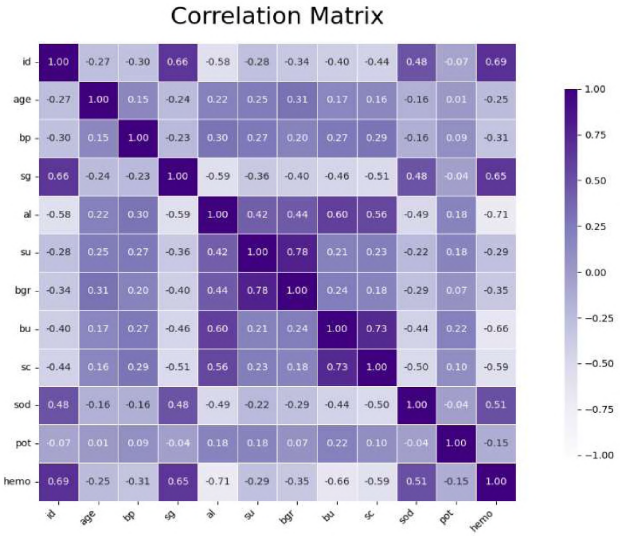


Fig.4. Correlation matrix

E. Machine Learning Models

In order to create a prediction technique, we try to combine the strengths of nine different machine learning models: Naive Bayes (NB), Logistic regression (LR), Decision Tree (DT), K-Nearest Neighbor (K-NN), Random Forest (RF), Support Vector Machines (SVM), AdaBoost (AB), CatBoost (CB), and Light Gradient Boosting Machine (LGBM). For combining ML models stacking algorithm is used.

III. Result and Discussion

A. System Requirements

As we really value our machine learning tasks, we also used a well-tuned system to provide the best performance and accuracy. Trimming the fat, shuffling and applying preprocessing on the data made sure to stabilize biases from training set; and a small adjustment was done by ensuring that random splitting after processing would not cause any data leakage. The main system with a solver: Intel® Xeon® 2-core CPU, 12.68 GB RAM, Tesla K80 GPU; fine-tuned for heavy loads Validation Verification is conducted on an Intel® Core™ i9- which ensures model robustness across different hardware. We were able to do this thanks to NumPy, Scikit-Learn and Pandas, three Python libraries which made the process more natural.

B. Confusion matrix

We assess the outcomes using a confusion matrix and classification report, taking into account memory, F1 score, accuracy, and precision. The model's training accuracy determines the output. The performance assessment indicators were identified using equations (3) through (6)

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using the percentages of True positives (TP), False positives (FP), True negatives (TN), and False negatives (FN) [18].

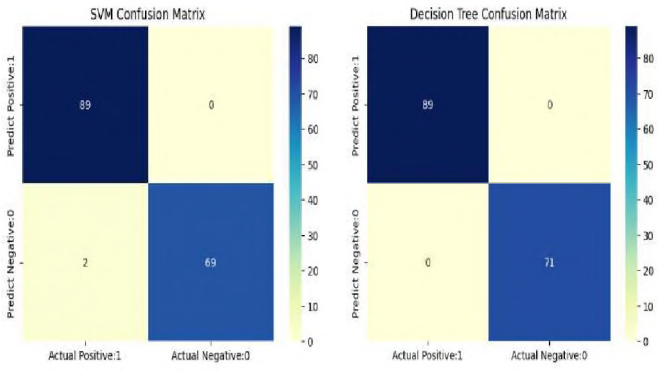
$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (3)$$

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

$$Recall = \frac{TP}{TP + FN} \quad (5)$$

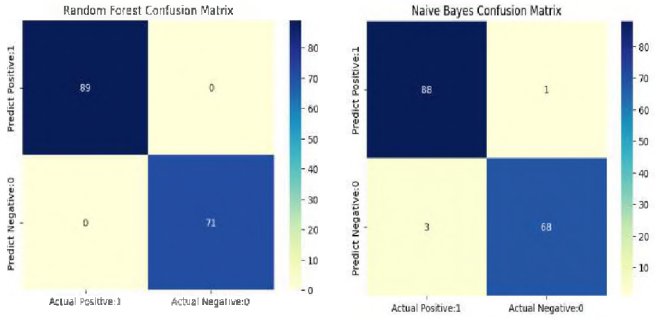
$$F1 = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (6)$$

The performance parameters of recall and precision are crucial in medical investigative research. When evaluating diagnostic performance, the F metric—which sums together recall and accuracy into a single, understandable figure—is quite helpful. Because false negatives can result in serious patient damage and undetected diseases, recall is frequently given precedence above precision in the medical industry. On the other hand, false positives are often less harmful since physicians can handle them, even though they may raise unwarranted fears. As a result, a modified F1 score that gives recall a higher priority than precision is a more useful indicator of diagnostic effectiveness. The confusion matrix for the machine learning approach employed in this investigation is shown in "Fig. 5". The associated matrix displays the number of True Positives, False Positives, True Negatives, and False Negatives for each model. In order to assess and compare each model's performance, these confusion matrices are essential. They include information on the accuracy, precision, recall, and general dependability of each model—all of which are crucial in applications involving medical diagnosis where incorrect classifications might have detrimental effects.



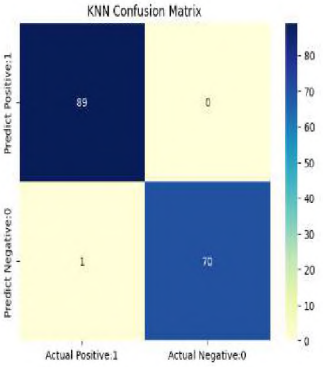
(a)SVM

(b) Decesion Tree

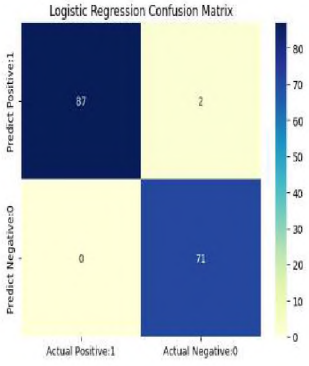


(c)Random Forest

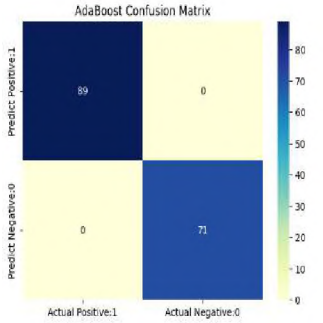
(d)Naïve Bayes



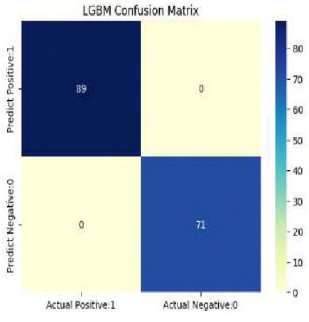
(e) KNN Classifier



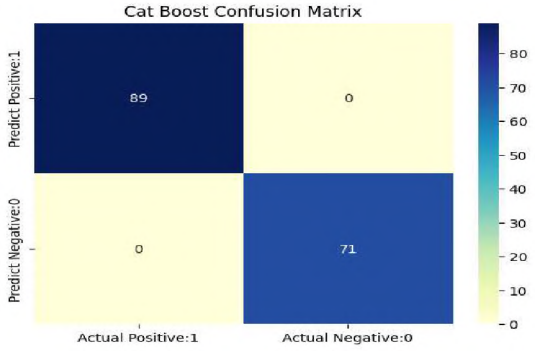
(f) Logistic Regression



(g)AdaBoost



(h) LGBM



(i)Cat Boost

Fig.5: Confusion Matrix of used dataset(a) SVM(b) Decesion Tree (c)Random Forest(d)Naïve Bayes (e) KNN Classifier (f)Logistic Regression(g)AdaBoost (h) LGBM(i)Cat Boost

Table I presents an accurate comparison between each model's training and testing data to help minimize overfitting. You may see the F1-score, Support, precision, recall, and accuracy of each model. ROC curves for several machine learning classifiers used to test data are shown in Fig.6.

TABLE I. PERFORMANCE EVALUATION OF SEVERAL ML MODELS

Name	Accuracy	Precision	Recall	F1 Score	Support
SVM	76%	76%	92%	83%	65%
DT	97%	98%	97%	98%	65%
RF	99%	100%	98%	99%	65%
NB	90%	100%	85%	92%	65%
KNN	89%	95%	88%	91%	65%
LR	96%	96%	98%	97%	65%
AdaBoost	99%	100%	98%	99%	65%
CatBoost	95%	98%	98%	98%	100%
LGBM	99%	100%	98%	99%	65%

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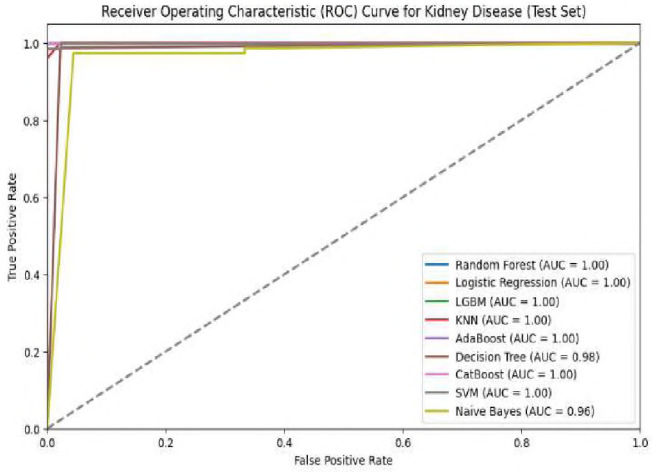


Fig.6. ROC curve for Test data

We show how each model works with ROC curves, which map out the True Positive Rate against the False Positive Rate at different cut-off points. ROC curves help us see how good a model is when the threshold changes. A key metric here is the area under the ROC curve, or AUC. The higher the AUC, the better the model does at telling things apart.

SGD learning was used to the inference to properly manage missing data, and binary formatting was carefully applied to classification variables. This procedure guaranteed the best possible data encapsulation and improved the performance of the model.

C. Stacking Ensemble

We analyzed chronic kidney disease diagnosis using a novel stacking ensemble approach. Our projections were far more accurate as a result of this. We combined Random Forest (RF), Decision Tree (DT), and Light Gradient Boosting Machine (LGBM) in a procedure called stacking. Each model makes a unique contribution: Decision Trees are easy to understand, Light Gradient Boosting Machine boosts it with its gradient tricks, and RFs stabilize it and avoid overfitting. Table II displays the efficacy of each combination. This illustrates how the combined model performs better than each one alone. We were able to generate quite outstanding results by integrating many models, including Random Forest Decision Trees, AdaBoost, Logistic Regression, and Light Gradient Boosting Machine. We achieved 99.19% accuracy, in fact. Every model has some advantages. We were able to reduce biases and provide robust, consistent predictions across several datasets with the use of this ensemble approach. Our findings demonstrate how stacking ensembles might enhance patient diagnostics and customize therapy regimens. Table II shows that complex algorithms and ensemble approaches do, in fact, increase forecast accuracy and dependability.

TABLE II. PERFORMANCE METRICS FOR CLASSIFIER COMBINATIONS OF ENSEMBLE

Combinations	Accuracy	Precision	Recall	F1 Score
LR + LGBM	97.33	0.99	0.99	0.99
RF + DT	99.19	0.99	0.99	0.99
RF + LR	99.17	0.99	0.99	0.99
RF + AdaBoost	99.16	0.99	0.99	0.99
LR + DT	98.67	0.99	0.99	0.99
LR+ DT+ LGBM	98.33	0.99	0.99	0.99
RF + LGBM	99.13	0.99	0.99	0.99
AdaBoost + LR	98.08	0.99	0.99	0.99
AdaBoost + LGBM	97.73	0.99	0.99	0.99
AdaBoost + DT	98.57	0.99	0.99	0.99
LR + DT + RF	97.75	0.99	0.99	0.99
LGBM + Ada + RF	98.88	0.99	1.00	0.99

Table II of the stack showcases results from various model combinations such as Logistic Regression + Decision Tree, Random Forest etc. Somewhere just below that we can see some scores like accuracy, precision, recall, F score etc. Those metrics that help us understand how well the different Strategies play together. Random Forest and Decision Tree together gave an outstanding accuracy of 99.19% And, well, everything else is not too shabby either (refer to table above). Because of the speed, it is one of the most preferred stacks in our analysis.

In this paper, we evaluate the performance of the proposed machine learning model in predicting CKC. We compared it with the newer methods researchers like K.M. Mohiuddin et al. [19], Qin et al. [20], Wang et al. [21], Mohammed et al. [22], and more. However, our model had used smart features to select from. These were combined with powerful ensemble techniques. By combining these we were able to achieve superb accuracy, precision and recall. For instance, Sara et al. [23] found accuracy rates of 90%. Conversely, our model reached 99.19% accuracy in the end. While our method had the best performance on F1 score and recall compared to most previous works. That demonstrates the improved prediction skill of our system due to the use of good selection of features. Table III confirms each accuracy for our technique outperforming most present approaches. We achieved 99.19% using Decision Tree (DT) and Random Forest (RF). Moreover, our model exhibits greater sensitivity in comparison to the other techniques previously mentioned.

TABLE III. COMPARISON TABLE WITH EXISTING WORK

Reference	Accuracy	Precision	Recall	F1 score
K.M. Mohiuddin et al. [19]	99%	99%	99%	99%
Qin et al. [20]	99.83%	_	99.80%	99.86%
Wang et al. [21]	76%			
Mohammed et al. [22]	99.17%	100%	98.68%	99.34%

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Sara et al. [23]	90%	95.24%	90.91%	93.02%
Dey et al. [24]	98%	99%	98%	99%
Our proposed model	99.19	99%	99%	99%

IV. Conclusion

Diagnosis of chronic kidney disease (CKD) requires considering a number of factors. The paper showed that the Random Forest and Decision Tree classifiers complement each other better than as standalone systems, in our case. Impressive, with a 99.19% accuracy rate. However, by mixing the two versions, it uses the best of each. It can help you to understand so much and make exact predictions, but a Decision Tree. The Random Forest, by contrast, uses loads of decision trees to avoid overfitting. This example of using this method for determining diseases also demonstrates an extraordinarily strong ability to control both variance and bias. In total, this method does the same by reinforcing mistakes from weaker models to get near perfect accuracy. In summary, stacking ensembles are a reliable manner to facilitate early detection of CKD. In line with the medical community's requirement for high sensitivity, our model prioritizes recall in our ensemble metrics, lowering the likelihood of undetected CKC occurrences. Medical experts may better understand prediction pathways thanks to the interpretability of the ensemble. This enhances the credibility and usefulness of clinical diagnoses.

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